Namal University, Mianwali

Project Proposal Document

Project: Point-of-Sale System

Course: CSC-225 – Software Engineering

Department: Computer Science

Semester: Fall 2025

Submitted To:

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Contents

1 Introduction

This document presents the system design for the Point-of-Sale (POS) System developed for a pizza shop as part of Project Milestone 3 of the course CSC–225 (Software Engineering). The objective of this milestone is to translate the approved Software Requirements Specification (SRS) into a complete and consistent system design.

The design strictly follows the functional and non-functional requirements defined in the SRS and adheres to IEEE 830 guidelines. Standard design artifacts such as use case, data flow, sequence, activity, class, and component diagrams are used to model system behavior and structure.

2 Design Assumptions and Constraints

2.1 Design Assumptions

- All users have basic familiarity with POS systems.
- Terminals operate within a Local Area Network (LAN).
- Reliable third-party payment gateways are available.
- Required hardware devices are available and functional.
- Multiple users can access the system concurrently.

2.2 Design Constraints

- Compliance with the approved SRS and IEEE 830 standards.
- Support for Windows 10+ and Android 10+ platforms.
- Real-time synchronization of orders.
- Secure role-based access control.
- Support for at least 50 concurrent users.

3 Key Design Decisions

3.1 Process Decomposition

The system is broken into smaller, manageable processes using hierarchical DFDs. This ensures that order placement, payment processing, and inventory management are modular and easy to maintain.

3.2 Order Lifecycle Management

A state-based order lifecycle was implemented to track orders from placement to serving. This decision ensures consistent order status updates, prevents duplication, and simplifies reporting.

3.3 Role-Based Access Control

Different user roles (Admin, Cashier, Kitchen Staff) are defined to control system access. This improves security and ensures that users can only perform actions relevant to their role.

3.4 Component-Based Architecture

The system is organized into modular components (UI, Auth, Order, Kitchen, Billing). This decision improves maintainability, allows independent testing of modules, and makes future scalability easier.

4 System Design Diagrams

4.1 Context Diagram

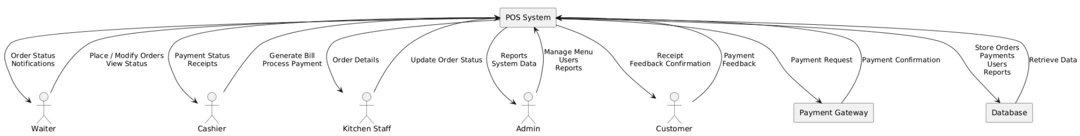
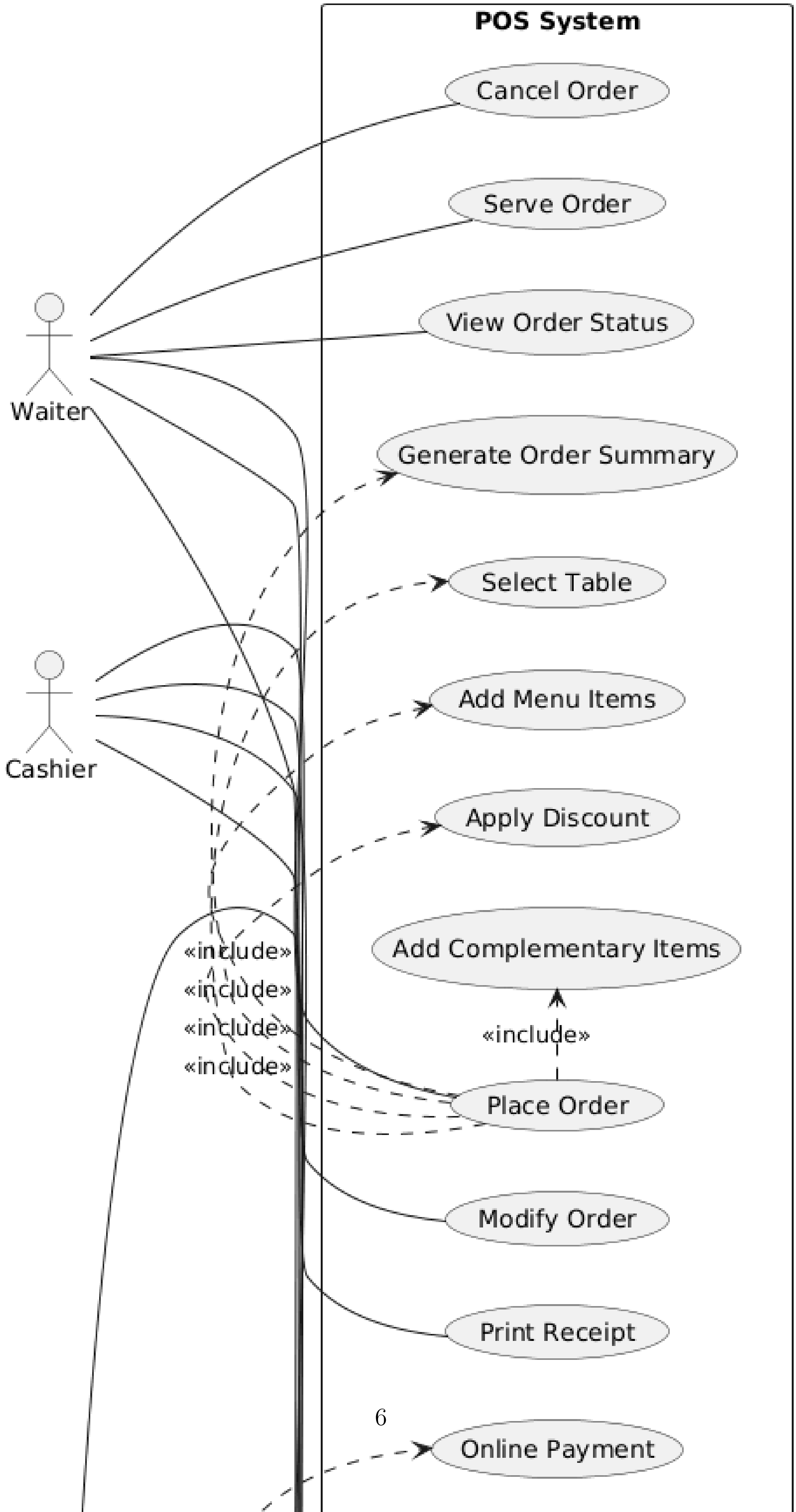


Figure 1: Context Diagram of the POS System

4.2 Use Case Diagram



4.3 Data Flow Diagram Level 0

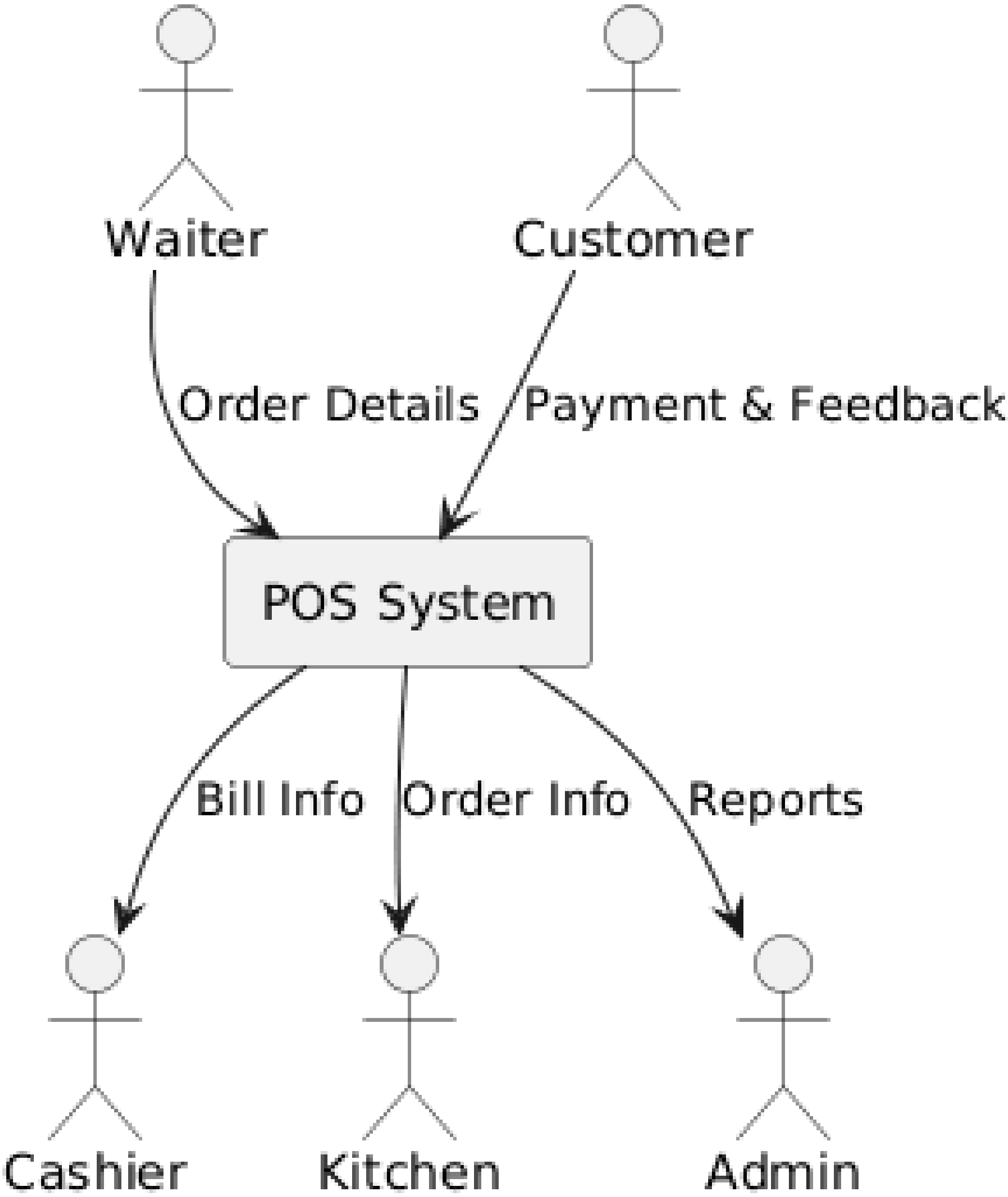


Figure 3: DFD Level 0 of the POS System

4.4 Data Flow Diagram Level 1

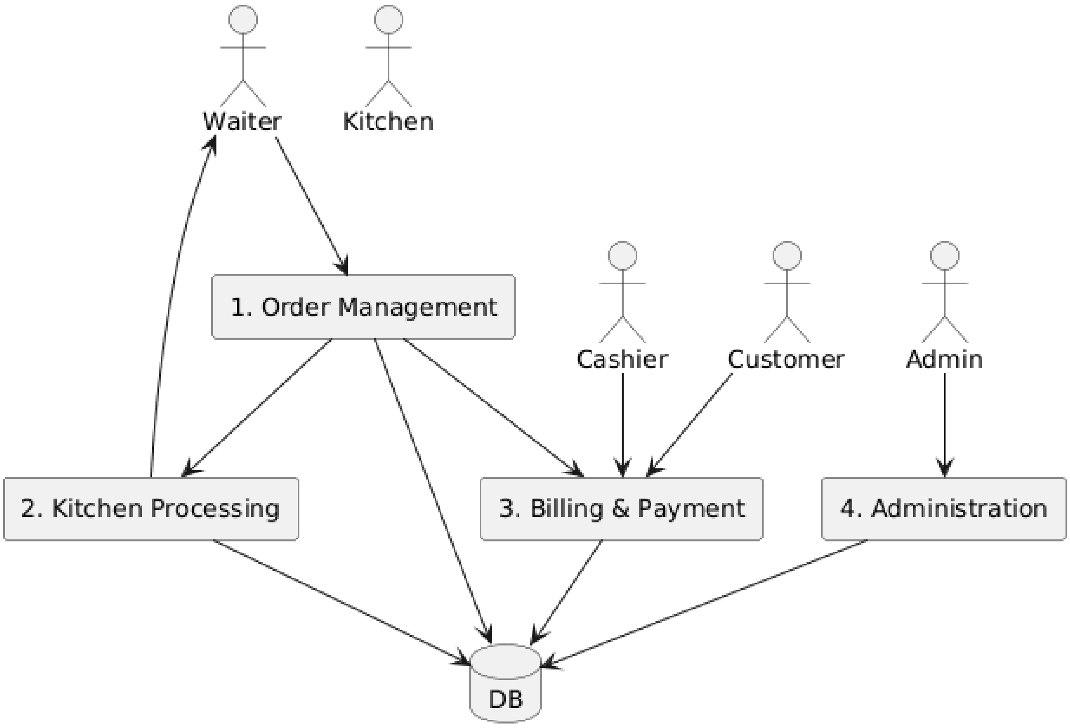


Figure 4: DFD Level 1 of the POS System

4.5 Data Flow Diagram Level 2

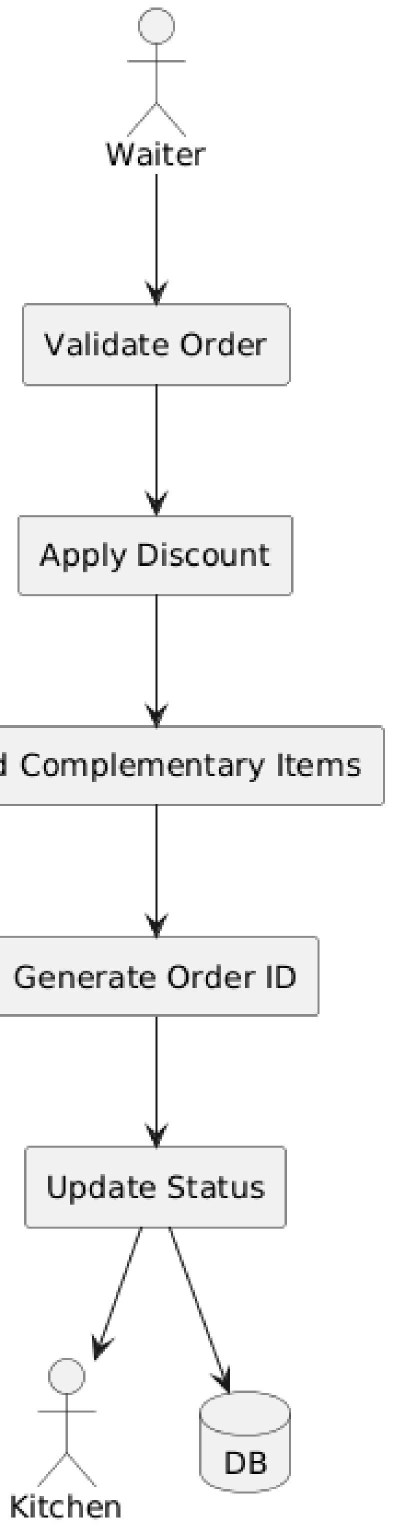


Figure 5: DFD Level 2 (Order Processing)

4.6 Sequence Diagram: Order Placement

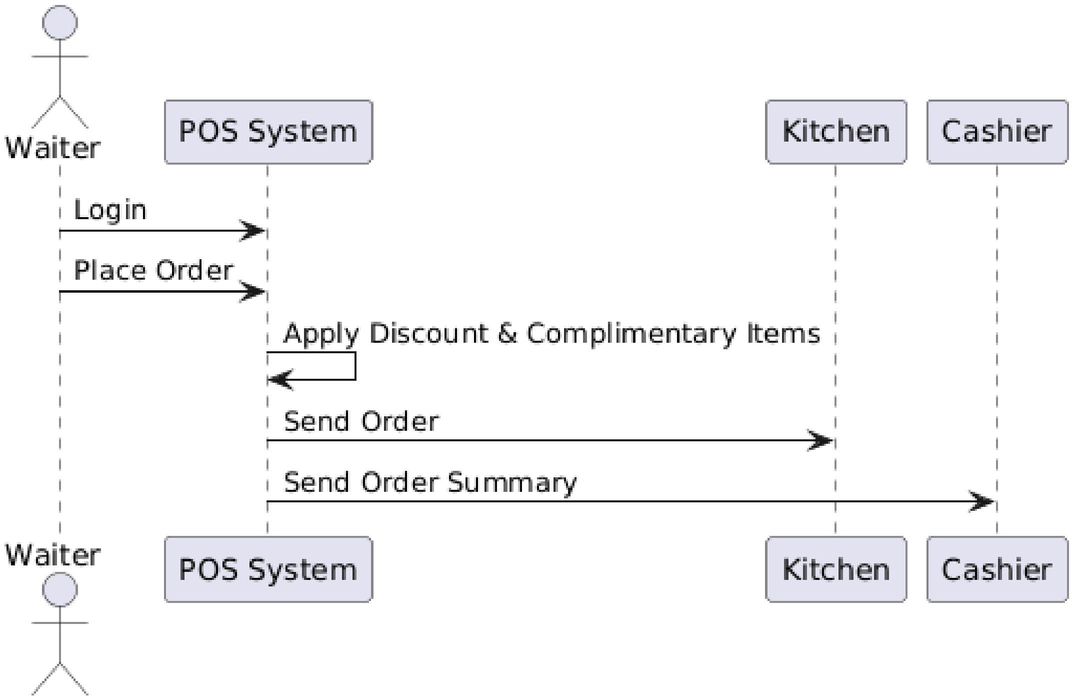


Figure 6: Sequence Diagram for Order Placement

4.7 Sequence Diagram: Payment Processing

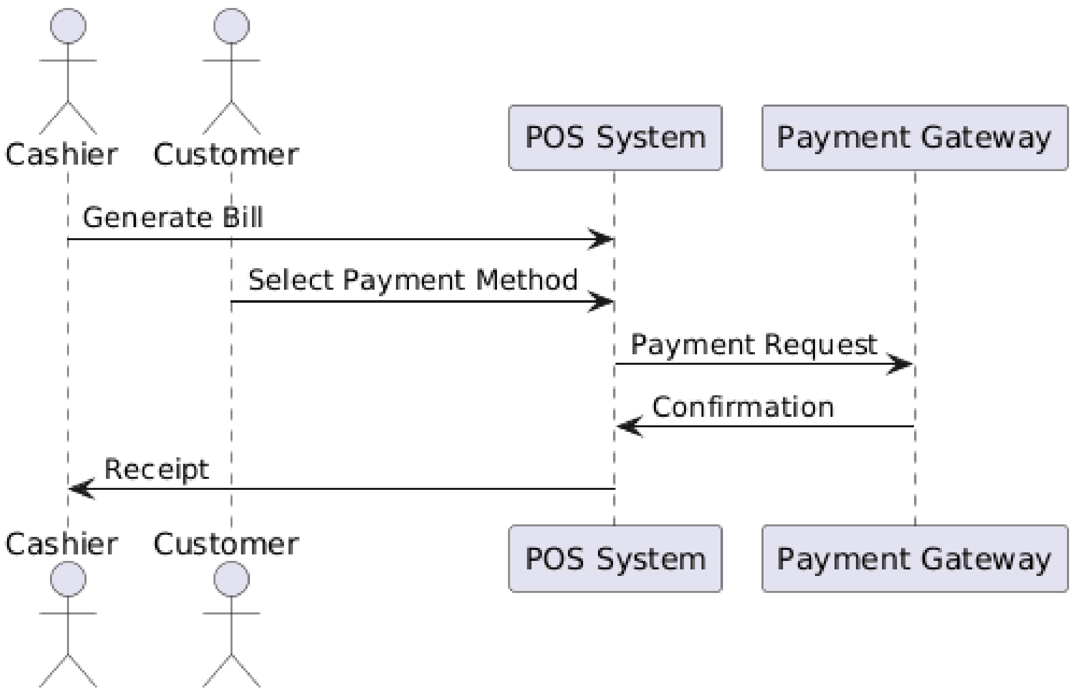


Figure 7: Sequence Diagram for Payment Processing

4.8 Activity Diagram

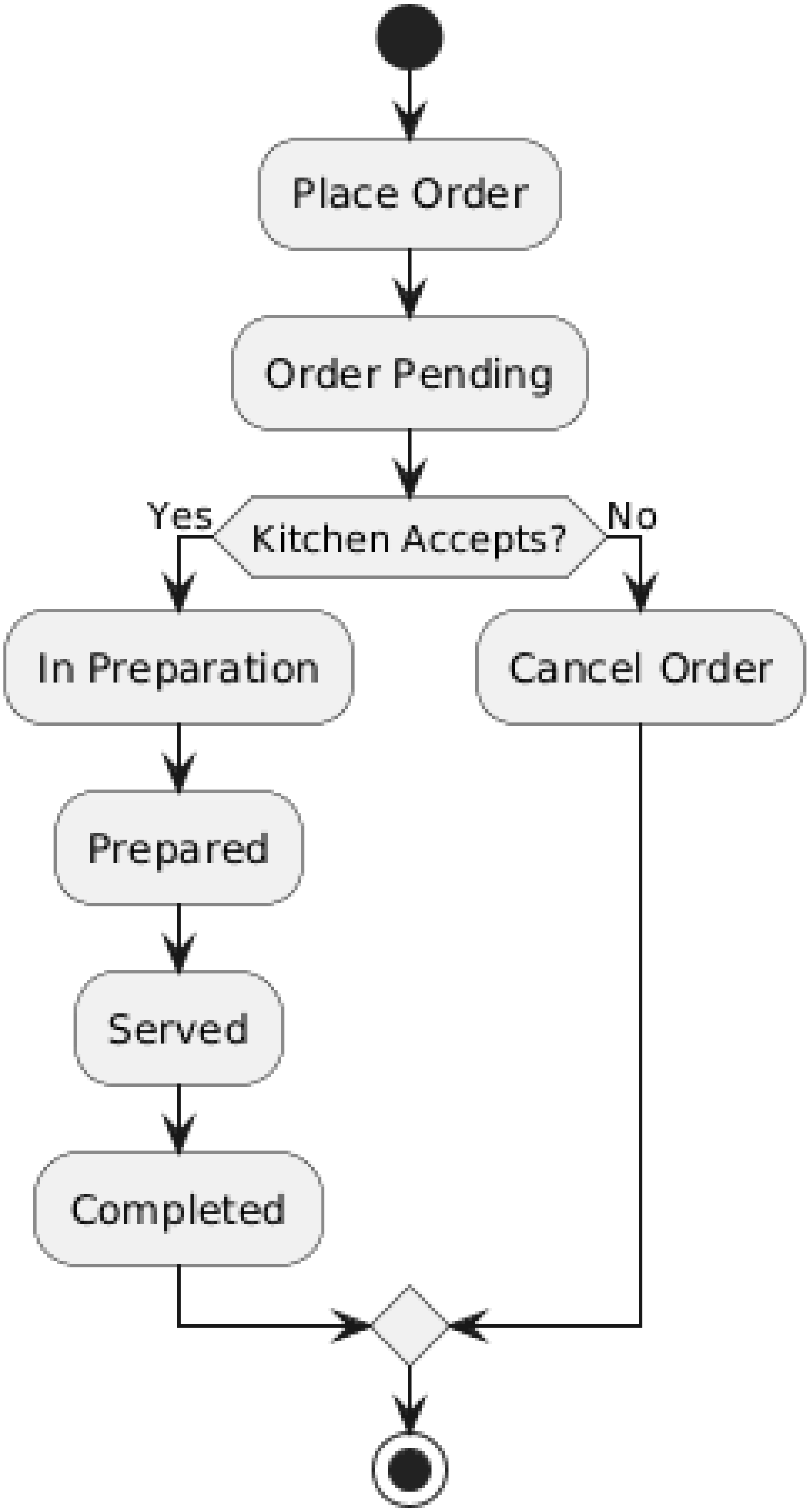


Figure 8: Activity Diagram of Order Lifecycle

4.9 Class Diagram 1

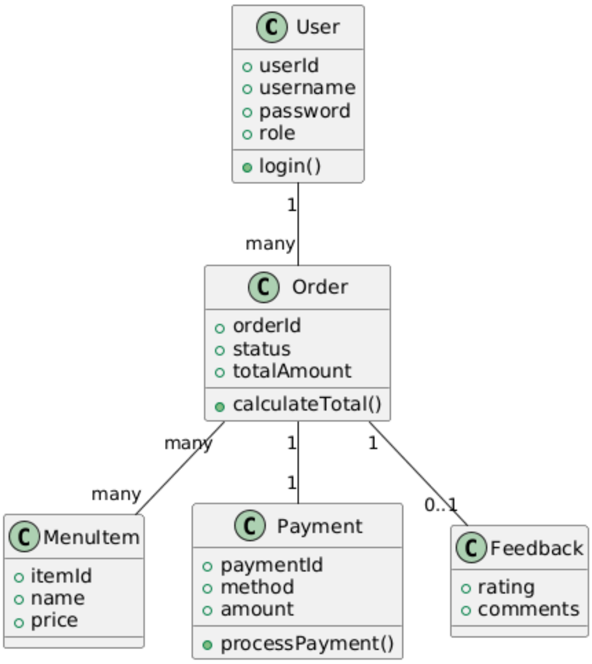


Figure 9: Class Diagram of the POS System

4.10 Class Diagram 2

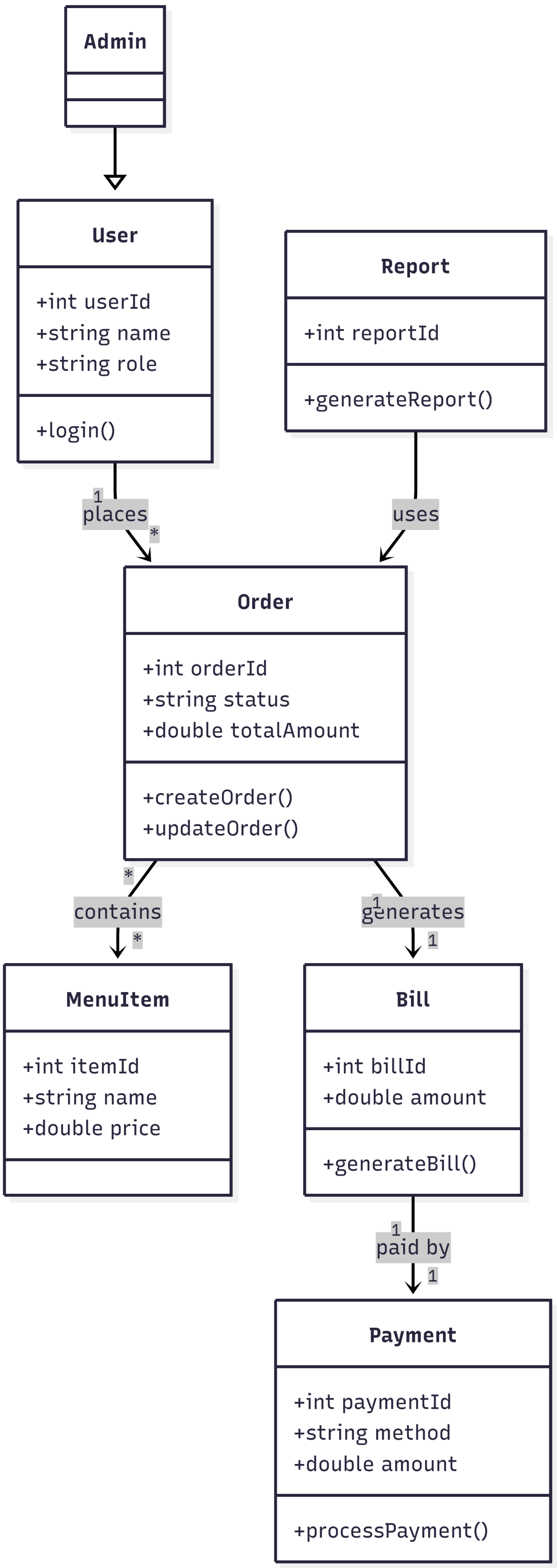


Figure 10: Class Diagram of the POS System

4.11 Component Diagram

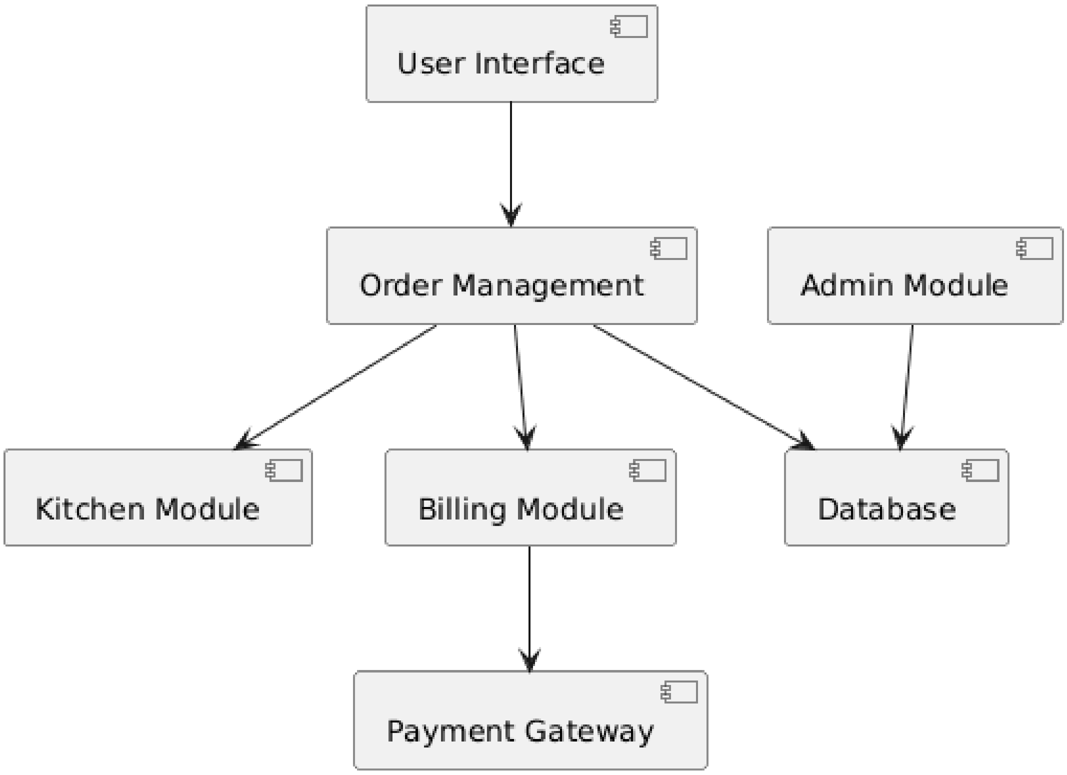


Figure 11: Component Diagram of the POS System

5 Requirements–Design Traceability Table

Req ID	**Description**	**Use Case**	**Design Artifacts**	**Components**
FR–01	User Authentication	Login	Use Case, Sequence	UI, Auth
FR–02	Order Placement	Place Order	DFD, Activity	Order Module
FR–03	Order Processing	Modify Order	DFD, Sequence	Order Module
FR–04	Kitchen Operations	Prepare Order	Sequence	Kitchen Module
FR–05	Order Serving	Serve Order	Activity	Order Module
FR–06	Billing & Payment	Process Payment	Sequence	Billing Module
FR–07	Feedback	Provide Feedback	Use Case	UI Module

FR–08	Administration	Manage System	Use Case	Admin Module

6 GitHub and Figma Links

- **GitHub Repository:** https://github.com/muhammadshoaibnoor/Software-Engineeting-Pr git
- **Figma Design Link:** https://www.figma.com/design/W9yBXlcReXCFpw3lnawDiS/Untitled?node-id=0-1&t=6qPDvcfS4w8IL03B-1

7 Conclusion

This report fulfills all requirements of Project Milestone 3. The system design is complete, IEEE-compliant, and fully traceable to the SRS.